MILITARY SPECIFICATION

CLEANER, PRESSURE, HOT-WATER, SKID-MOUNTED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of a portable, skid-mounted, electrically-operated, oil fired, hot water pressure cleaner, with a minimum output capacity of 200 gallons per hour (gph) at a minimum 1,000 pounds per square inch gage (psig) operating pressure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

VV-F-800 - Fuel Oil, Diesel.
VV-F-815 - Fuel Oil, Burner.
ZZ-H-601 - Hose and Hose Assemblies, Rubber (Wrapped or Braided) Water Service.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronics and Associated Equipment).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-E-16298 - Electric Machines Having Rotating Parts and Association Repair Parts: Packaging of.

MIL-C-25769 - Cleaning Compound, Aircraft Surface, Alkaline Waterbase.
MIL-H-28596 - Hose Assembly, Rubber, Wire Reinforced, 200 PSIG, Saturated Steam Service.

STANDARDS

FEDERAL

Fed. Std. H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A53 - Welded and Seamless Steel Pipe.
A106 - Seamless Carbon Steel Pipe for High Temperature Service.
D2156 - Smoke Density in Flue Gases from Distillated Fuels, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS - General Standards for Industrial Controls and Systems.
MG1 - Motors and Generators.
WD1 - General Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 296 - Oil Burners.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cleaners shall consist essentially of an oil-fired, continuous coil, spark-ignited heating unit; and electric-motor-driven, positive-displacement water pump; a concentrated cleaning solution storage and dispensing system; operational and safety controls; a skid-mounting; a spray gun with nozzles; and hoses specified herein.

3.2 First article. When specified (see 6.2.1), the contractor shall furnish one complete cleaner for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. The cleaner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the cleaner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Performance.

3.6.1 Capacity. The cleaner shall be capable of pumping a minimum of 200 gph of water through the heating coil, raising the temperature of the liquid up to 180° Fahrenheit (F) by adjusting a temperature control, and delivering the water to the high pressure water gun at a minimum pressure of 1,000 psig.

3.6.2 Vibration and noise. The cleaner shall operate without excessive or undesirable noise, vibration, or pulsations. Valves shall close without causing water hammer.

3.7 Design. The cleaner shall be designed and constructed to clean dirty grease, oil, and mud from earth-moving and other equipment used in Military construction projects and shall be capable of withstanding the normal shocks and stresses incident to transport by truck, air, sea, and rail. The design shall permit accessibility for maintenance and service of cleaners in the field and shall be such as to prevent conditions which may be hazardous to operating personnel or deleterious to the equipment. The cleaner shall be designed and equipped to permit complete drainage of the water flow system, including the heating coil, pump(s) and piping. Water and solution container shall be so designed and located as to prevent spillage or drainage on electrical equipment. All screw threads shall conform to Fed. Std. H28.

3.7.1 Cleaning solution selection. The cleaner shall upon demand, deliver under pump pressure either concentrated hot or cold water solution of cleaning compound, or a mixture of cleaning compound and water, or water. The cleaner shall be suitable for use with a 25-percent water solution of cleaning compound conforming to MIL-C-25769.

3.8 Safety. All mechanical or electrical parts that are of such nature or so located as to become a hazard to operating or maintenance personnel shall be enclosed or properly guarded. The cleaner shall comply with applicable OSHA Standards, Title 29, CFR, Chapter XVII, Part 1910.

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3.8.1 Safety controls. The cleaner shall be equipped with all mechanical and electrical controls and devices necessary for safe and efficient operation. Such controls shall, as a minimum, include the following:

a. Safety control to shut down the burner in the event the temperature of the heating coil exceeds a predetermined safe limit due to partial or total interruption of the water flow to the coil. This control may be responsive to coil temperature, coil expansion, water flow rate, water storage levels, or any equally reliable indicator, at the option of the manufacturer.
b. Mechanical or electrical pressure control device to prevent the pressure in the heating coil and delivery hose from exceeding a predetermined safe limit.
c. Automatic fuel shut-down device in event of ignition interruption.
d. A safety-relief valve suitable for relieving water.

3.9 Heater section. The heater section shall consist essentially of a heating coil, a combustion chamber, and an oil burner. The burner shall be capable of continuous operation at full rated capacity with a density reading not to exceed ASTM D2156 (Shell Bacharach Scale) No. 2 spot for No. 2 fuel oil.

3.9.1 Heating coil. The heating coil shall be of the continuous, tubular type. The coil shall be fabricated of seamless or welded carbon or alloy steel pipe or boiler tube having sufficient ductility for close-coiling. The coil, after being formed, shall be capable of withstanding a hydrostatic test pressure of 2,000 psig, except that coils formed of seamless or electric-resistance welded pipe conforming to ASTM A53 or ASTM A106 may be tested at the applicable mill test pressure specified therein. The nominal diameter of the tubing shall be at least 1/2-inch. The cleaner shall be equipped with accessible connections, fittings, and valves, as required, to permit coil blowdown for the removal of scale deposits and sludge. The coil shall be designed and installed to allow for unequal expansion and contraction and to facilitate removal for soot-cleaning, repair, or replacement. The coil shall be enclosed in an insulated sheet-metal jacket or casing. When required to prevent conditions hazardous to operating personnel, a secondary jacket to provide an annular space for circulation of cooling air around the primary casing shall be furnished.

3.9.2 Combustion chamber. The combustion chamber shall be formed of preformed castable refractory; of a double, concentrically-coiled, tangent-tube water wall; of a preformed, ceramic-fiber refractory; or of a tangent-tube water wall with metal liner. Both the castable and ceramic-fiber refractories shall be rated to withstand exposure of the hot face to temperatures up to 2,500°F.

3.9.3 Burner. The oil burner shall be of the high-pressure mechanical atomizing type or the low-pressure air atomizing type. The burner shall have a rate thermal input sufficient to meet the performance requirements of 3.6. The fuel system for mechanical atomizing burners shall include a positive-displacement, electric-motor-driven fuel pump. Low pressure burners shall include a fuel metering device. Both type burners shall be equipped with high-voltage electrode-type spark ignition and a forced draft blower. The burner firing cycle shall be thermostatically controlled in accordance with the manufacturer's standard practice to prevent excessive heating coil temperatures at reduced steam-cleaning-cycle flow rates. Burner shall conform to UL296. The burner shall be suitable for firing any of the following fuel:

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a. Fuel oil, grade No 2 or VV-F-815 (API gravity: 30.0 minimum).
b. Diesel fuel, grades DF-1 and DF-2 of VV-F-800 (API gravity: 38.0 to 45.0).
c. Jet fuel (kerosene type), grade JP-5 of MIL-T-5624 (API gravity: 36.0 to 48.0).

The cleaner shall be equipped with a fuel tank having a minimum capacity for 2 hours continuous operation.

3.10 Water pump. The cleaner shall be equipped with a positive-displacement, electric-motor-driven water pump of the double-diaphragm type, or the packed plunger, duplex type. The pumping system shall be designed and rated to provide the flow rates and pressures specified in 3.6 herein. Pump check valves, including disk or balls, springs, seats, and any other associated metallic parts, shall be fabricated of stainless steel, copper alloys, nickel alloys, or other alloys equally resistant to the corrosive effects of the steam cleaning compound, and coil-descaling acid solutions. Pump bearings shall be of the sealed, permanently-lubricated type or shall run in an oil bath.

3.10.1 Pulsation dampener. The pump or piping connected thereto shall be equipped with a damping device of the air chamber (accumulator) type, spring-loaded diaphragm type, or equivalent type to absorb pressure pulsations caused by the reciprocating action of the pump.

3.10.2 Water float tank. A tank with a constant level float valve shall be provided to act as a reservoir for the incoming water supply. The tank shall be piped to the water pump inlet connections. The tank shall be fabricated of corrosion-resistant metal or metal treated to resist corrosion, and shall be equipped with inlet piping consisting of a manual shutoff valve, and a standard threaded connection for a 3/4-inch garden hose coupling. The inlet float valve shall be designed to close against supply pressures up to 80 psig. The tank shall be equipped with an overflow pipe and an air gap or other means to prevent back-siphonage of the tank water into the water supply line.

3.10.3 Water supply hose. One 3/4-inch by 50 foot length of hose for feeding water to the water float tank shall be furnished. The hose shall be yarn-reinforced, rubber construction conforming to grade 3, class 1, size 3/4 inch of ZZ-H-601. The hose shall be equipped with commercial brass couplings one end female, the other male or female, as required, to mate with the water tank fitting.

3.11 Cleaning solution system. The system for introducing concentrated cleaning solution into the water system shall consist essentially of a storage container, dip tube, and a manually adjustable metering valve or device. The container shall be equipped with means for filling and for drainage. The metering valve shall have a minimum selective range of 0 to 5 gph. The solution container shall be resistant to the corrosive effects of the cleaning solution and shall have a capacity of at least 10 gallons. The dip tube shall be properly sized and shall be of sufficient length to reach to bottom of the container.

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3.12 Delivery equipment. Each cleaner shall be furnished with a cleaning gun, a set of nozzles, and 40 feet of high pressure hose. The cleaner shall be equipped with a hanger, rack, or other appurtenance for stowing the guns and hose in transit, and when otherwise not in use.

3.12.1 Cleaning gun. The gun shall be furnished as a single unit with interchangeable nozzles. The gun shall be equipped with an insulated or air-cooled, trigger shutoff, pistol grip configured handle. The wand shall be a tube not less than 2 feet in length. The handle shall be capable of withstanding a 2,000 psig hydrostatic test. The handle and wand shall be capable of withstanding the drop test specified in 4.4.1.7.

3.12.2 Spray nozzle. Three corrosion-resistant steel nozzles shall be furnished with each cleaner to provide 0, 15, and 40° spray patterns. When specified (see 6.2.1), spare spray nozzles shall be provided with the unit. The nozzles shall be equipped with quick-disconnect couplers and shall not leak during cleaner operation. The nozzles shall be drop resistant, and shall be prominently marked so as to assist the cleaner operator in selecting a particular spray pattern.

3.12.3 Hose. The hose shall be suitable for delivery of high-pressure hot water. The hose shall have a manufacturer's working pressure rating of not less than 2,250 psig, and all hose fittings shall be reusable. Hose shall be 3/8-inch inside diameter nominal with a length of not less than 40 feet. The hose shall consist of a wire-braid reinforced, natural or synthetic rubber, multi-ply inner carcass, and a heat-abrasion, oil and weather-resistant, neoprene or equivalent outer cover. Hose shall be similar to MIL-H-28596.

3.13 Electrical system. The electrical system shall be comprised of motors, selector switches, sensors, and wiring hardware required to operate the system under the specific operating conditions. The power source shall be 208-volt, 3-phase, 60 Hertz.

3.13.1 Motor. The motor shall be designed and rated in accordance with NEMA MG1. Motors shall be furnished in dripproof enclosures and sealed windings. Motor overload protection, load capacity, feeder circuit protection, controller selection, and associated wiring practices shall conform to the requirements of NFPA No. 70. Motor controller shall meet the design and operational requirements of NEMA ICS

3.13.2 Wiring. Wiring on the cleaner shall be complete up to the point of the plug connection for the power supply cord. All wiring and circuit devices shall conform to, and be installed in accordance with, NFPA No. 70.

3.13.3 Power cord. A 75-foot hard-service, 4-conductor, oil-resistant, flexible, power cord shall be furnished with each cleaner. The cord shall be NEMA type S, SO, or ST and shall be equipped with 3-phase, 3-pole, 4-wire grounding type connectors conforming to figures 15-20 of NEMA WD1. One end shall be male, the other female. The cleaner shall be equipped with a matching, weather-proof male connector suitable for exterior applications.

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3.14 Enclosure. Components of the cleaner requiring protection from the weather, such as the pumps, motors, switches, and other controls, shall be enclosed in a weather-resistant, 20-gage or heavier, sheet-metal housing with access doors or panels and a stock cover. All switches, and to the maximum practicable extent, operating handwheels for the manually adjusted metering valve and control valves shall be centrally located in an orderly-arranged control panel. The control panel shall be recessed or otherwise guarded to prevent damage to protruding elements. Access doors or panels shall be equipped with suitable latching devices.

3.15 Skid mounting. The cleaner shall be mounted on a welded, structural steel skid base. The skid base shall include two longitudinal structural steel skid runners and a base plate or welded cross members to support the cleaner. When the cleaner is supported on cross members, open areas between the skids shall be covered by a sheet steel splash pan not less than 0.0598-inch (16 gage) thick. The skid runners shall be suitable I-beams, channels, or equivalent shapes. The runners shall be of sufficient height to provide ready access to any connections located under the units and of sufficient width to provide adequate bearing surface on a smooth gravel base when the cleaner is full-serviced for operation. In no case shall the clearance from the ground to the lowest projection of the cleaner be less than 4 inches. Both ends of each skid shall be shaped to facilitate skidding and shall be provided with holes or attachments for inserting, towing, and lifting slings. The skid runners shall be provided with pockets to permit lifting of the cleaner by a forklift truck. The pockets shall be a minimum of 24 inches on centers and shall be shaped to accommodate forks in sizes up to 2 inches in thickness by 8 inches in width.

3.16 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.17 Instruction plates. The cleaner shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

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a. Safety precautions.
b. Preparation of the cleaning solution.
c. Descaling procedures and schedule for the heating coil.
d. Blowdown procedures.
e. Wiring diagram.

The following shall also be permanently and legibly stenciled on the equipment in letters at least 3/4-inch high in conspicuous locations:

a. Use only No. 2 fuel oil; diesel fuel, or JP-5.

All switches, and control valves shall be identified by permanent plates marked in a manner specified herein for the metallic instruction plates.

3.18 Nameplate. A nameplate will be furnished by the contracting officer for each unit. A contractor shall stamp all necessary data in the blank spaces provided for that purpose and securely affix it to each unit in a conspicuous place with nonferrous screws, bolts, or rivets, not less than 1/8 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank. The contractor's standard metal nameplate will be acceptable in lieu of the Government-furnished plate, provided the plate, as a minimum, is marked with the manufacturer's name, model number, serial number, and electrical data.

3.19 Technical manuals. Technical manuals pertinent to cleaner operation, maintenance, and spare parts shall be furnished in accordance with the applicable provisions of the contract (see 6.3)

3.20 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.21 Electromagnetic interference control. When specified (see 6.2.1), the units shall meet the electromagnetic interference control requirements and test limits for class C3 equipment as specified in MIL-STD-461.

3.22 Workmanship. Workmanship shall be of the quality prevalent among manufacturers normally producing high pressure cleaners used in truck and bus fleet maintenance shops, for construction equipment, and aircraft cleaning operations. The quality of workmanship shall not adversely affect safety, function, performance, serviceability, interchangeability, and appearance. Workmanship shall not exhibit defective material or processing such as, seams, laps, laminations, cracks, fins, extraneous material, visible steps or irregularities, sharp edges, nicks, scratches, burrs, tool scores and gouges, deformations, missing operations, improper assembly, missing parts, stains, corrosion, nonspecified oxidation (rust), unauthorized salvaging operations (e.g., hammering to shape, repair by welding, straightening, bending, etc.,). Unless otherwise specified, for electrical, electronic parts, and assemblies, and workmanship provisions of MIL-STD-454 applies.

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3.22.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.23.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one complete cleaner when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.3 and the tests of 4.4.1 and, when required, the test of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.2, and the packaging inspection of 4.6.

4.3 Examination. Each cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 First article tests. When first article is required, the first article shall be tested as specified in 4.4.1.2 through 4.4.1.6 The first article shall be performed by the contractor under the direction of the Government representatives. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all cleaners until corrective action has been taken. Acceptance of first article shall not constitute a waiver by the Government of the right under the provisions of the contract.

4.4.1.2 Hydrostatic test. Each heating coil, after assembly and prior to installation, shall be hydrostatically tested at the pressure specified in 3.9.1 for a period of at least 2 minutes after the coil is fully prepared and under test pressure. Any coil exhibiting leakage or rupture shall be rejected.

4.4.1.3 Functional test. Each cleaner shall be operated for at least 1 hour on a high-pressure, hot water wash cycle to verify the functional adequacy of operational components and controls. Any cleaner which requires frequent adjustment; suffers flame failure or erratic combustion; fails to maintain the specified operational pressures; or otherwise exhibits nuisance shutdowns or repeated malfunctions shall be rejected. All normal minor adjustments shall be effected during the test run to insure that the cleaner is in acceptable operating condition prior to delivery.

4.4.1.4 Performance test. The first article sample, when furnished, shall be tested to verify compliance with the requirements of 3.6 and 3.9. The duration of the test shall be 2 hours for a high pressure hot-water wash cycle. The following data shall be recorded at 20 minute intervals during the test:

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a. Liquid flow rate in pound per minute (recorded as a meter reading or scale reading for calculation of flow rate after test).
b. Temperature of water leaving coil.
c. Water pressure at coil outlet.

Data recorded during the test shall be used to verify compliance with the following requirements of this specification:

a. Flow rates.
b. Discharge pressures.
c. Discharger temperatures.

The solution feed system shall be checked during the test for compliance with 3.11.

4.4.1.5 Safety control tests. After the test of 4.4.1.4, the first article sample, when furnished, shall be tested to verify compliance with 3.8. The conditions which activate the safety controls shall be simulated by interruption of water and fuel flows, as applicable, and for the pressure control test, by throttling the discharge.

4.4.1.6 Drop tests. The cleaner's handle with wand and delivery hose-assembly attached shall be dropped two times from a height not less than 6 feet onto a smooth concrete floor. One drop shall impact on the handle and one drop shall impact on the wand. Upon completion of the two drops, the handles and wand shall be inspected for permanent deformation, breaks, and fractures. Either breakage or fracture of the handle and wand shall be cause for rejection. Permanent deformation of the handle and wand shall not be cause for rejection. The cleaner shall then be operated, and the handles and wand inspected for leaks. Leaks correctable by tightening of threaded connections shall not be cause for rejection. Leaks that cannot be corrected by tightening of threaded connections shall be cause for rejection. Failure of the cleaner to operate satisfactorily shall be cause for rejection.

4.4.2 Quality conformance test. Each production cleaner shall be operated and tested as specified in 4.4.1.3.

4.5 Electromagnetic interference control tests. When electromagnetic interference control is specified (see 3.21), the first article shall be tested in accordance with MIL-STD-462 to verify compliance with the applicable provisions of MIL-STD-461.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, and commercial as specified (see 6.2.1).

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5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.6 Unit protection. Each complete cleaner shall be preserved method IIa. Openings in the equipment shall not be sealed, so that desicated air may circulate.

5.1.1.7 Burners, combustion chambers, and fuel tanks. Unpainted ferrous metal surfaces of the burners, combustion chambers, and fuel tanks shall be coated with type P-10 preservative.

5.1.1.8 Heating coils, solution tanks, and pumps. Interior surfaces of the heating coils, solution tanks, pumps, piping, and fittings for the water system shall be coated with type P-3 or P-21 preservative. Excess preservative shall be drained.

5.1.1.9 Power cord. The power cord shall be coiled to a minimum safe diameter and secured.

5.1.1.10 Hoses. The water hoses shall be coiled to a minimum safe diameter and secured.

5.1.1.11 Spray gun and nozzles. Interior surfaces of spray guns and nozzles requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-3 or P-21 preservative. Excess preservative shall be drained. The spray gun assembly shall be either secured in the compartment provided in a manner to prevent damage and movement during shipment or, if disassembled, shall be preserved method IC.

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5.1.1.12 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.13 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.14 Consolidation. If provided, tool boxes or compartments shall be utilized for the consolidation of tools, accessories, and technical publications. The remainder of components that will not fit in the tool boxes or compartments shall be consolidated in boxes conforming to PPP-B-636, class domestic. Contents shall be cushioned, blocked, and braced to prevent movement within the container. The consolidation container shall be secured to the equipment and shall be within the method II barrier.

5.1.2 Level B. Preservation shall be as specified for level A with the following modifications:

5.1.2.1 Unit protection. Each complete burner shall be preserved method I.

5.1.2.2 Electrical motors. Preserve level A by the alternate method of MIL-E-16298.

5.1.2.3 Consolidation. Consolidation shall be as specified for level A except that containers shall be class weather-resistant.

5.1.3 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. Each complete cleaner shall be packed in a box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be anchored, blocked, and braced to prevent movement within the container.

5.2.2 Level B. Each complete cleaner shall be packed as specified for level A except the boxes shall be domestic type or class 1

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cleaner covered by this specification is used for cleaning dirt, mud, grease, and oil from automotive equipment, heavy construction equipment, ground support equipment, and miscellaneous parts for maintenance and overhaul, the cleaning is to be performed by the application of high-pressure hot or cold water washes or rinses.

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6.2 Ordering data.

6.2.1 Acquisition documents should specify the following:

a. Number, title, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
c. If spare nozzles are required (see 3.12.2).
d. Finish coat color required (see 3.16).
e. When fungus resistance treatment is required (see 3.20).
f. When electromagnetic interference control is required (see 3.21).
g. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.19	Publications, Commercial	DI-M-24010A	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the cleaner will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1). The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing activity:

Navy - YD

Project No. 4940-N455

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